

Locating a Drug Runner: Miami Vice Style

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Summary

We have been assigned to develop a procedure to determine accurately the region in which a suspected drug runner is located, and to find the “optimal” helicopter search method to capture the runner. We make initial assumptions regarding the specifics of the helicopter and the powerboat, the accuracy of the detection devices, the motions of the drug runner, and the specifics of the search. The probable region containing the speedboat is determined by applying statistical analysis to the accuracy of the detection devices. We investigate several possible search strategies, and the elimination of ineffective patterns results in the consideration of two promising paths. We test our model based on drug-runner psychology and the two search strategies. The development of a computer simulation of the interaction between the helicopter and the speedboat allows us to test variations in each scenario. We discuss the resulting data and present an optimal solution based on our model. The model is evaluated, and we suggest further improvements and refinements.

Assumptions and Hypotheses

The problem statement contains several unknowns that we will have to assume; however, these assumptions are either based on documented information or appear reasonable in all cases.

Helicopter Specifics

We assume that the helicopter is similar in design and specifications to the U.S. Coast Guard Sikorsky HH-3E, which are well able to reach a

speed of over 200 mph [Boyne and Lopez 1984, 108, 249]. However, we assume that the helicopter's speed is a constant 180 mph (264 ft/s), since the pilot must be actively searching the ocean below. We also assume that the helicopter's searchlight is mounted underneath the cockpit, so that a full 360° "sweep zone" is possible. We further restrict this zone to a 90° region extending 45° on either side of the forward motion of the helicopter. This refinement is reasonable if we assume that the searchers can view only a region directly ahead of their path. Since we reduce the sweep zone by three-quarters, we will assume that the searchlight can illuminate this entire region continuously; that is, the beam can sweep fast enough to illuminate the entire 90° region. Since the weather is calm, we assume that the helicopter can fly safely within 20 ft of the ocean. Finally, we assume that the helicopter is sophisticated enough to allow an exact flight pattern to be followed and maintained. This seems acceptable if we assume that the helicopter has a computerized navigation system.

Speedboat Specifics

Since the problem statement specifies the drug runner's powerboat has a speed of one-third the helicopter, we assume that the boat speed is 60 mph. Furthermore, we assume that the runner's acceleration is instantaneous and that the boat maintains top speed when moving. Since the typical "go-fast" motorboats often used by drug runners can obtain speeds of 80 mph [Shannon 1986], these assumptions appear reasonable.

Detector Accuracy

We interpret the error of $\pm 2^\circ$ as a normal distribution, centered on the given angles (119° and 110°), with standard deviation of 1° . Thus, the true values will fall within 2° of the measured value with probability of approximately 95%.

Psychology of the Drug Runner

We assume that the drug runner will remain at the original position indefinitely, as long as the helicopter is not heard. If the runner does hear the helicopter, we assume that the boat will attempt to escape by following a path perpendicular to the direction of motion of the helicopter and will move at maximum velocity. The runner's motion following the detection of the helicopter is a factor that we consider separately.

Specifics of the Search

Considering limitations in fuel and pilot fatigue, we assume a 6-hr time limit on the search. That is, if the runner is not sighted after 6 hrs, the search will end and the runner will have escaped. Certainly, the drug deal would have been completed within this time limitation. We also assume that the helicopter engages the search at a specified point by flying from the coast at an altitude of 600 ft, beyond the hearing range of the runner, and then descending to water level. In this manner, the helicopter may begin the search at any arbitrary point while maintaining the same starting conditions. Finally, if the powerboat is within sight of the helicopter, we assume that the drug runner will be apprehended.

Brief Problem Overview

The first problem to attack is the description of the region in which the pilot can expect to find the waiting powerboat. The given information includes only the results of the sensing devices used to triangulate the possible locations of the drug runner. Since we assume that the sensing devices operate according to a prescribed statistical method, we must use some sort of probability interpretation to describe accurately the region in which the boat will be found. In any case, however, we can take the given accuracy of the sensing devices, $\pm 2^\circ$, to calculate a first approximation of the desired region. This result will give the *minimum possible* region in which the boat lies waiting and is instructive in further analysis.

After determining the correct region in which the speedboat lies, the next problem is to devise an optimal search strategy to capture the drug runner within the assumed 6-hr time limit. We interpret “optimal strategy” as the method that will give the highest success rate in capturing the speedboat. This interpretation disregards flight time, gas expenditure, and ease in implementation; however, we feel that the best strategy should concentrate on the actual *capture* of the drug runner, regardless of any other factor.

The most important consideration in developing a search strategy is the fact that the speedboat can hear the helicopter at 500 ft, but the pilot can see the boat at only 200 ft. Thus, we have the case that any helicopter approach pattern has the highly probable result of “scaring” the speedboat away from the pickup point, without a helicopter sighting. This consideration led us to consider only “closed-curve” flight patterns; that is, ones in which the helicopter would follow a circulating path. As a result, the helicopter has higher probability to discover the drug runner if the helicopter follows an inwardly or outwardly progressing path. We will investigate two such flight patterns: perfect circles centered on the most probable location of the speedboat, and quadrilaterals following the general dimensions of the region in which the pilot expects to find the drug runner.

Another important consideration is the movement of the speedboat once the drug runner hears the helicopter. The responses could be quite diverse; however, we concluded that three such responses are most probable:

- The runner might simply continue fleeing, never going back to the pickup point, thinking that the drug deal has been scrubbed.
- The boat might move away from the pickup point, but after a certain time stop and wait indefinitely for further instructions from its contact.
- The boat might initially move away from the point, but after a certain time stop and head back to the same point to complete the drug exchange.

Each of these three scenarios certainly has strengths and weaknesses; but without more detailed information on the landscape and other factors, we include all three for completeness.

Problem Analysis

Defining the Region

Since we are assuming 95% confidence that the sensing devices used to locate the speedboat are accurate to 2° , we may use the *interval error method* to determine the extent of the possible location of the waiting speedboat. That is, we may examine the maximum and minimum deviations in each sensor to determine the shape and size of the resulting region. The result of this basic trigonometric procedure are shown in **Figure 1** and magnified in **Figure 2**. Although the region is quite extended in two directions, we note that the *most probable* location of the drug runner is at the intersection of the 110° and 119° sensor lines from the listening posts. This is a result of our assumption that the true angle varies on a normal distribution centered at the given angle. Thus, for further computations and analysis, we choose to relocate our origin from the second listening post to the point of greatest probability density (see **Figure 3**).

Developing Optimal Search Pattern

Central to the problem is the idea of the “catch zone,” the region about the helicopter in which the runner will be caught. The problem states that at an altitude of 200 ft, the helicopter’s searchlight will illuminate a region 25 ft in radius. However, if the helicopter maintains an altitude of about 20 ft, the effective search zone increases to a radius of almost 200 ft (see **Figure 4**). Thus, any object within 200 ft of the helicopter and within the beam of the searchlight will be detected. Essentially, then, except for the initial “drop in” at the start of the search, we have reduced the problem to two dimensions.

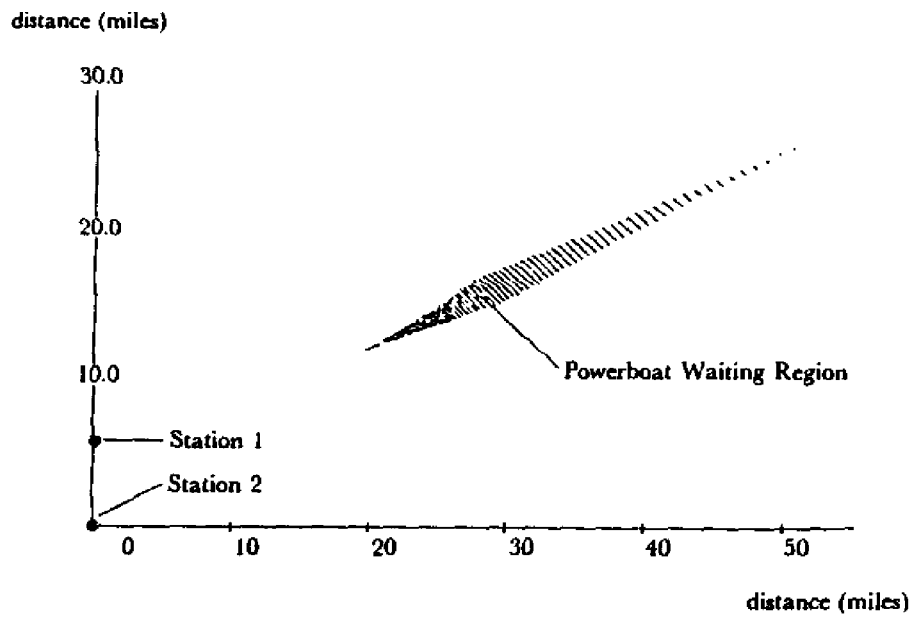


Figure 1. The powerboat waiting region as determined by the interval error method.

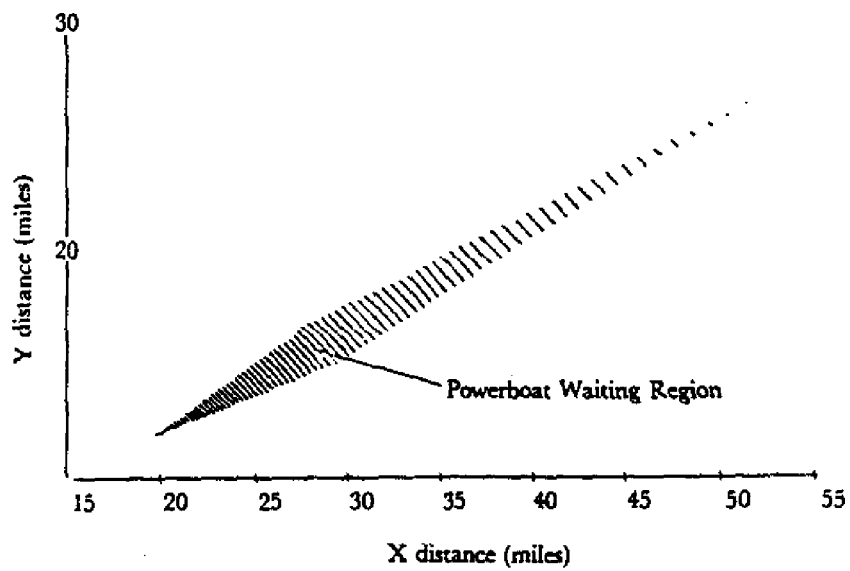


Figure 2. Magnification of Figure 1.

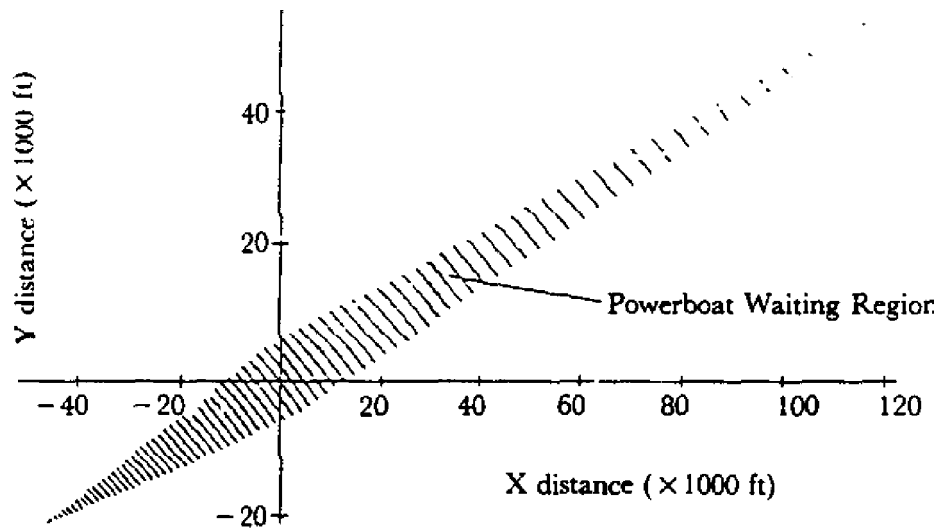


Figure 3. As in Figure 2, but with origin relocated to point of greatest probability density.

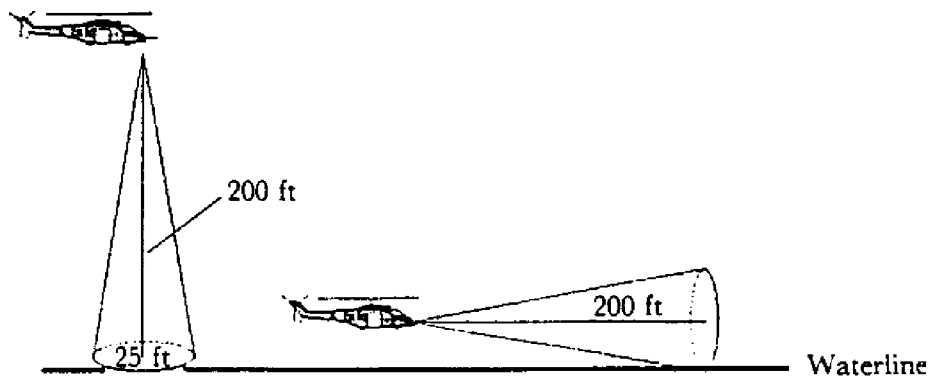


Figure 4. Different ways to use the searchlight.

The searchlight is allowed to sweep out a continuous path 90° wide, creating a “sweep zone” that represents the extent of the searchers’ vision (see Figure 5).

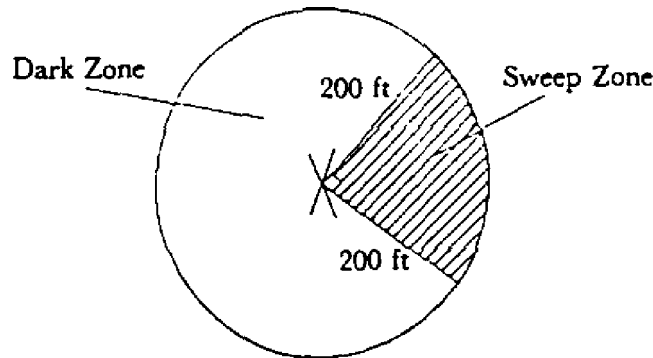


Figure 5. Extent of the searchers’ vision.

The remaining portion of the 200-ft-radius circle, the “dark zone,” represents that portion *not* instantly swept out by the beam. However, since the helicopter is flying forward at 264 ft/s (180 mph), this dark zone will in effect be searched, since the helicopter has already moved through the region. Thus, we have a “catch region” 283 ft across, which guarantees the drug runner’s capture if the speedboat lies within the region (see Figure 6).

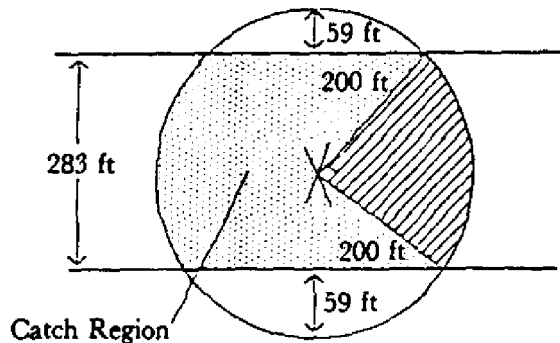


Figure 6. Geometry of catch region.

Although there are two additional 59-ft portions of the pilot’s 200-ft radius outside the catch region, we cannot envision a situation in which the boat would be within these sections, since the drug runner has a 300-ft hearing advantage and would have moved out of the helicopter’s range. Effectively, then, if the speedboat lies anywhere within a 200-ft radius of the helicopter, then either the drug runner is in the 90° sweep zone and will be apprehended, or else has already been captured.

An important factor in developing the search pattern is the difference between the drug runner’s 500-ft hearing range and the helicopter’s 200-ft

sight zone. It is likely that on any given pass, the drug runner will both hear the helicopter and be outside the catch region, thus initiating the boat's movement away from the helicopter's path. Our search pattern, then, must take into account the probable result that the boat will be moving in some fashion. We initially considered many search patterns but found that several would not be feasible. One possibility was some sort of spiral, such as the spiral of Archimedes ($r = a\theta$) or the exponential spiral ($r = ae^{\theta}$), which continually curve away from the center. Unfortunately, any such spiral became impractical due to difficulties in establishing parametric equations (as a function of time) that would allow constant velocity along the length of the spiral. Other possibilities included any set of straight line segments or sine waves adjusted to the boundaries of the search region. These proved impractical, however, since initial calculations showed that it is impossible to cover the *entire* region within the 6-hr time limit. Thus, the best strategy would be to follow a path that would both cover the region of highest probability (the origin) and also "loop" back upon itself, increasing the likelihood of intercepting the boat.

The two patterns that we examine are shown in **Figures 7** and **8**, superimposed on the powerboat waiting region.

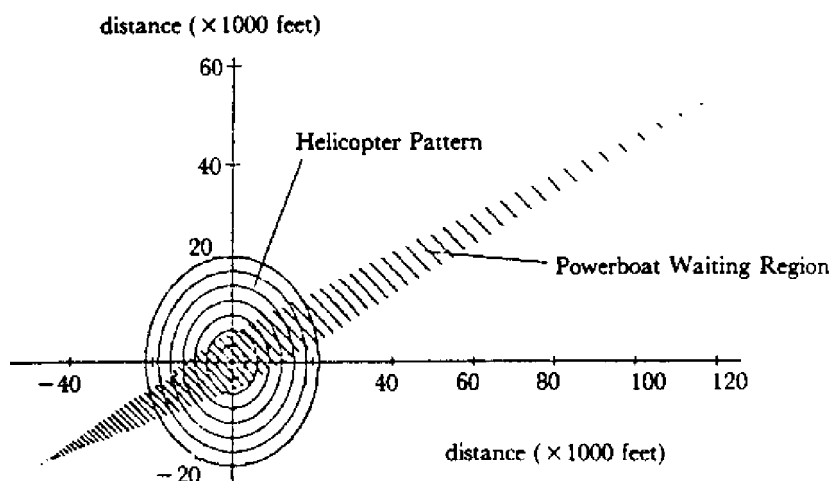


Figure 7. Circular Method of search.

In the Circular Method of **Figure 7**, the helicopter travels along circles centered at the most probable point in the region. In the Quadrilateral Method of **Figure 8**, the helicopter travels along a pattern geometrically similar to the boundaries of the powerboat waiting region. We will investigate the difference in capture rates for varying distances between successive patterns and varying starting points. We will also examine the effects of changing the direction of travel ("inward" or "outward") along the given pattern. (Note that the distances between successive patterns is quite large

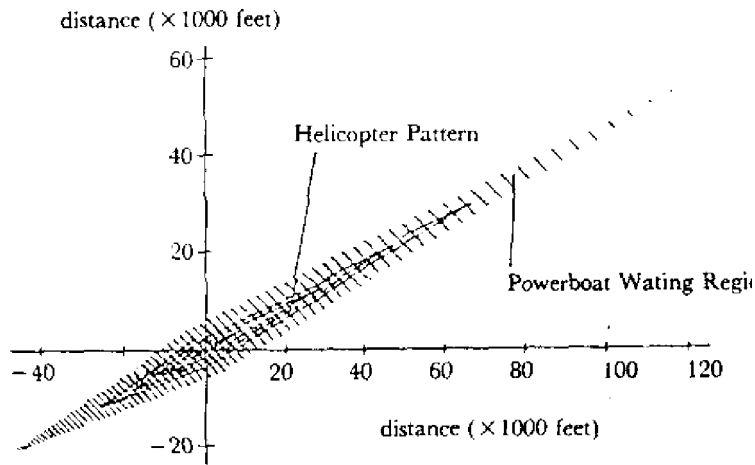


Figure 8. Quadrilateral Method of search.

in the figures; this is for illustration only, since for the practical search one would want no more than perhaps 2,000 ft between successive patterns.) Notice that in the circular pattern, the helicopter travels significantly beyond the limits of the powerboat waiting region, while in the quadrilateral scheme it remains entirely within the region. This difference will likely manifest itself in the three different drug-runner reactions described above, in that the Circular Method will likely perform best for the drug runner's flee response, while the Quadrilateral Method will be best for the cases in which the runner stops and waits or returns to the pickup point.

Model Design

The heart of our model is a set of computer routines that simulate the interaction between the helicopter and the speedboat according to the assumptions stated previously. The computer routines are flexible enough to allow us to investigate a variety of different factors in the interaction and perform repeated application of these factors in a large number of test cases.

Our computer routines reduce to a $3 \times 2 \times 4 \times 2$ model, in which we examine the effects of the following four factors on the capture rate:

- *Drug runner's response to hearing the helicopter.* The runner is assumed to exhibit one of the following three responses:
 - The boat moves away from the sound of the helicopter and neither stops nor returns to the initial point.
 - The boat initially moves away from the sound but stops 5 mi away and waits for further instructions.

- The boat moves away from the sound, stops 5 mi away, but returns to the initial pickup point after waiting 10 min, provided the runner does not hear the helicopter again.
- *The type of flight path followed by the helicopter.* We include consideration of the two types of “closed paths” mentioned above, the circular and quadrilateral schemes. In either method, the helicopter will navigate complete loops.
 - In the Circular Method, the helicopter begins and ends each circle at a specified distance along the x -axis, and then travels directly along the axis before beginning the next circle (see **Figure 7**).
 - In the Quadrilateral Method, the pilot begins and ends each loop at a specified distance along a line joining the most probable point (the origin) and the closest exterior corner point of the powerboat waiting region to the origin (see **Figure 8**).
- *The distance between successive loops in either the Quadrilateral Method or the Circular Method.* This parameter effectively indicates the “tightness” of the search pattern: For a large distance between each loop, the helicopter covers a larger total region but misses a significant portion of the intervening area. For a small distance, on the other hand, the helicopter covers a small region more thoroughly. Considering the helicopter’s 200-ft sweep zone, we examine the success rate of four different distances: 200, 500, 1,000, and 2,000 ft.
- *The radial direction of the helicopter’s path:* that is, whether the pilot follows successive loops outward from or inward toward the most probable point. Logically, the helicopter would start either at the most probable point and work outward or at an outer extremity and progress inward, since the pilot would want to “scare” the drug runner in the same direction.

Therefore, our analysis requires the simulation of 48 different configurations, from which we will draw conclusions as to the most successful method under various constraints.

The computer program is an interactive routine that uses basic algebraic methods to compute the flight path of the helicopter at one-second intervals throughout the 6-hr period. The program then determines if an interaction between the helicopter and the speedboat occurs at any point, subsequently calculating the motion of the drug runner according to the criteria established above. The process is simplistic, certainly; but it establishes an acceptable approximation to an actual search routine.

For each of the 48 configurations, we tested the computer simulation for the same 134 starting locations of the powerboat. These 134 points were derived in the following manner. We generated random numbers according to a normal distribution centered about the two given angles, 110° and 119° ,

and having a standard deviation of 1° , based on the assumption that the given error of 2° represents a 95% confidence interval. Triangulation of these pairs of random angles yielded 134 starting locations within the powerboat waiting region (see **Figure 9**).

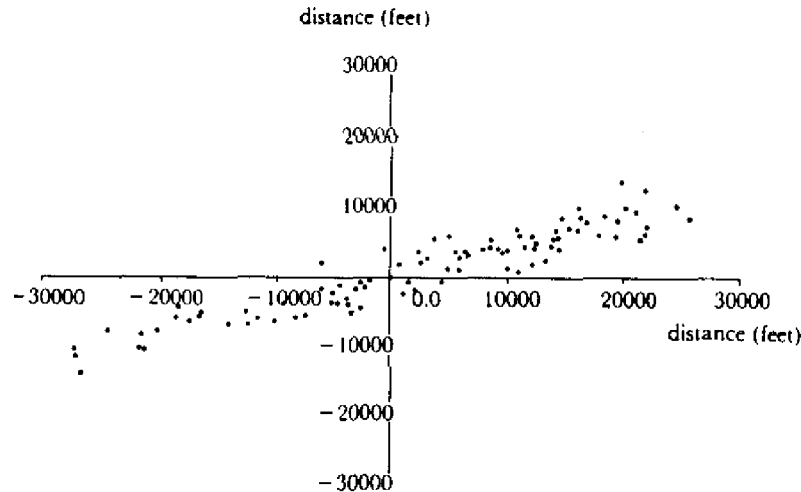


Figure 9. Search locations within the powerboat waiting region.

The sample size of 134 will result in a p (probability of catching the powerboat) for each of the 48 scenarios which will be accurate to within .051 at a 95% confidence level, assuming a p in the neighborhood of .10:

$$z_{.025} \sqrt{\frac{p(1-p)}{n}} = 1.96 \sqrt{\frac{(.10) \cdot (1-.10)}{134}} = .051.$$

Problem Predictions and Results

We choose to examine the capture rate based on the three possible “psychologies” of the drug runner. That is, for each of the responses—either stopping, returning, or fleeing after the initial sound of the helicopter—we investigate the best possible search strategy. We feel that this method is most practical, since various factors such as weather and coast configuration will alter the runner’s response. Using this scheme, then, our results can be applied for a variety of runner responses.

Tables 1–3 show the success rate—that is, the number of captures for the 134 initial boat locations—for the various parameters discussed above and for each drug-runner response. For example, in **Table 1**, for the Circular Method, starting at a large radius and progressing inward in steps of 500 ft, we find that the drug runner was captured in 18 of the 134 starting points, giving a success rate of .134.

Table 1.

Strategy 1: Drug runner flees pickup point and doesn't stop or return.
Number of catches (proportion of catches) for 134 random starting points.

Patterns:		Distance between concentric patterns			
		200 ft	500 ft	1,000 ft	2,000 ft
Circular	Outward	1 (.007)	14 (.104)	19 (.142)	11 (.082)
	Inward	2 (.015)	18 (.134)	22 (.164)	10 (.075)
Quadrilateral	Outward	8 (.059)	12 (.090)	7 (.052)	5 (.037)
	Inward	10 (.075)	9 (.067)	14 (.104)	14 (.104)

Table 2.

Strategy 2: Drug runner flees pickup point and stops to wait.
Number of catches (proportion of catches) for 134 random starting points.

Patterns:		Distance between concentric patterns			
		200 ft	500 ft	1,000 ft	2,000 ft
Circular	Outward	1 (.007)	13 (.097)	14 (.104)	11 (.082)
	Inward	6 (.045)	25 (.187)	23 (.172)	9 (.067)
Quadrilateral	Outward	12 (.090)	11 (.082)	7 (.052)	4 (.030)
	Inward	10 (.075)	9 (.067)	11 (.082)	12 (.090)

Table 3.

Strategy 3: Drug runner flees pickup point, stops to wait, and returns.
Number of catches (percentage of catches) for 134 random starting points.

Patterns:		Distance between concentric patterns			
		200 ft	500 ft	1,000 ft	2,000 ft
Circular	Outward	7 (.052)	18 (.134)	18 (.134)	12 (.090)
	Inward	15 (.112)	23 (.172)	21 (.157)	9 (.067)
Quadrilateral	Outward	12 (.090)	9 (.067)	7 (.052)	3 (.022)
	Inward	10 (.075)	11 (.082)	9 (.067)	14 (.104)

These tables suggest that the optimal strategy of those tested is a circular pattern with inward movement and intermediate distance (500 or 1,000 ft) between consecutive rings. This is not surprising, since a circular pattern increases the probability that the helicopter will intercept the boat should the runner have already been set in motion. An inward, as opposed to outward, pattern will tend to chase the boat toward the center of the region, rather than “out to sea,” thus effectively reducing the region in which it is most likely to be caught. Patterns with small distances between each ring will cover only a small portion of the total region in the allotted time, while very large patterns leave a greater space in which the boat may be passed over. Thus, those of intermediate size should result in a higher capture ratio, as observed.

However, as the tables show, there are some exceptions to these general rules. While the Circular Method tends to be superior in most cases, at a very small distance between concentric rings, the Quadrilateral Method results in a higher catch ratio. Further, the superiority of the inward direction seen so clearly with the circular pattern is not realized when the quadrilateral search pattern is used.

The particular cell that shows the highest capture rate varies from chart to chart, so it is difficult to make a specific recommendation to the helicopter pilot. The best course of action depends on the strategy selected by the drug runner, a strategy that the pilot could not know. If we assume that the three strategies that we investigated occur with equal probability, then the probabilities of a capture are equivalent for the Circular-Inward-500 strategy and the Circular-Inward-1,000 strategy. It is also important to consider the statistical error of approximately 0.05 based on our random sample size, so choice of a clear-cut optimal strategy is further clouded.

If we consider the perspective of the drug runner, then Strategy 3, in which the boat returns to the initial spot to complete the drug deal, is by far the poorest; it is most likely to result in capture. From this point of view, there is no appreciable difference between the other two strategies.

While movement in the inward direction appears optimal in terms of our criteria—the proportion of trials that result in capture within 6 hrs—this may not be true in terms of other criteria. Because the region of highest probability of finding the drug runner is reached last in this pattern, a greater expenditure in terms of effort, helicopter fuel, etc., may be required on those trials that do result in capture.

Model Strengths and Weaknesses

The most important technical weakness in our procedure lies in our consideration of only two helicopter flight patterns. An infinite number of paths could be investigated, some quite complex. We believe, however,

that our examination of the two methods (circular and quadrilateral patterns) provides a good indication of the model's success. Smoothing the circular pattern out into a true spiral would probably be more efficient, but the parametrization proved impractical. The model also is limited in its assumption about the drug runner's path following the initial sound of the helicopter. While we assume an escape path perpendicular to the tangent of the helicopter's direction, in fact the boat's motion could be quite chaotic. Similarly, our analysis considers only three drug-runner responses (flees, waits, or returns). The real-life response could be considerably more complex, perhaps even combining elements of all three. Also, we do not consider the time of flight of the helicopter from the Coast Guard station to its initial search point. This seems acceptable, however, since even the most distant starting point would be less than 31 mi from Post 1, a distance that the helicopter could cover in less than 15 min, a small portion of the total search time. Finally, our definition of "optimal" neglects the time factor involved in the search; we treat only whether the runner is caught at *any* time within the 6-hr limit. If it became necessary to minimize the search time, further analysis would be necessary.

The most important strength of our model lies in the calculation of the exact trajectories of the helicopter and the speedboat at *each second* within the 6-hr time interval. The computer routine therefore examines 21,600 points of possible helicopter-speedboat interaction, eliminating the possibility that the paths will cross undetected. Our model is successful in that it considers many possible scenarios. We examine 48 different configurations, varying flight patterns, directions, and drug runner responses. Our computer routine may be conveniently adapted to include consideration of any other helicopter flight pattern. This flexibility would facilitate future changes or improvements necessary in the scenario. Finally, the model is statistically accurate, since the initial powerboat points are chosen according to the more realistic assumption that the accuracy of the sensing devices is represented by a normal as opposed to a uniform distribution.

References

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- Shannon, E. 1986. Running silent, running fast. *Newsweek* (27 October 1986): 94–95.